INCH-POUND

A-A-54746
25 September 1992

COMMERCIAL ITEM DESCRIPTION

# STOPCOCK, INTRAVENOUS THERAPY, 3-WAY, PLASTIC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-S-37434 which is cancelled.

This Commercial Item Description covers a sterile, disposable, stopcock with three separate fluid paths and attached tubing.

Salient characteristics:

Material. The stopcock and male Luer connector shall be fabricated from a rigid or semi-rigid plastic. The rotating spigot shall be of a color contrasting to the stopcock body.

The plastic tubing shall be fabricated from transparent, pliant plastic. Tubing shall be flexible, but shall be rigid enough to prevent collapse during normal use.

The plastic shall be inert to blood and common IV fluids and meet the requirements of the USP for Class II plastics. The total combined quantity of barium, cadmium, and lead for each component (excluding end protectors) shall not exceed 5 parts per million (ppm).

Style, design, and dimensions. The stopcock set shall be of the styles shown in accordance with Figure 1:

Style 1: w/20 inch extension tubing

Style 2: w/33 inch extension tubing

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The set shall consist of a stopcock with a rotating spigot and a secured length of tubing to the stopcock body. A male Luer connector (needle adapter) shall be secured to the distal end of the tubing. The injection site shall be assembled to the rotating spigot. The injection site shall enable piercing and immediate resealing characteristics when punctured with needles of 18 gage or smaller.

The stopcock proper shall have two side ports, 180° apart, each having internal female Luer tapers. The cavity of the rotating spigot that contains the rubber injection site shall also possess a female Luer taper. Luer fittings shall be in accordance with ISO Standard 594 and may be female Luer Lock. The rubber plug of the injection site shall be attached by means of a retainer which allows for removal of the injection site. Once removed, the rubber plug shall not be reattached. The rotating spigot shall have an arrow-type handle to indicate which female side Luer port will be part of the fluid path.

The two stopcock open ports and male Luer connector shall be covered with readily removable plastic protectors. The end protectors shall maintain cleanliness and freedom from particle infiltration and shall maintain sterility of the fluid paths until manually removed.

Performance. The stopcock set shall be a unit suitable for use in administering the direction of flow of blood and solutions for simultaneous and/or alternate intravenous therapy. Stopcocks shall be tested for freedom from leakage, clear passage of fluid, and security of assembly.

Storage. The stopcock set shall be tested to ensure freedom from tackiness, brittleness, or other evidence of deterioration when stored in an elevated temperature.

Sterility. The stopcock set shall be sterile in accordance with USP or AAMI guidelines. Sterility Assurance Level shall be .000001.

Instructions for use. A detailed instruction sheet for proper use of the stopcocks shall be provided with each individual stopcock or with each unit package.

Workmanship. The Luer tapers and rotating spigot shall be free from parting lines. The stopcock shall be free from any defects which affect its appearance or utility.



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Unit. Package (PG). One package containing fifty individually packaged sterile stopcocks, with instructions, as specified, constitutes one unit. The individual package shall be a sealed, peel-open type container capable of maintaining sterility of contents unless the package is damaged or opened.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, with an inspection level of S-4. Unit of product for test purposes shall be one set. A test failure shall be cause for rejection.



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Clear passage of fluid. The stopcock set shall be tested with an appropriate testing fluid applied internally to the stopcock body with the stopcock body placed in a horizontal position. Each fluid path as shown in Figure 2 shall be tested, in turn, using testing assembly fixtures in accordance with ISO Standard 594 except that the male connector shall not be plugged and that the female Luer cavity of the rotating spigot shall be tested by removal of the rubber injection site. The set shall allow free passage of the testing fluid within five (5) seconds after an internal pressure of 3 psi maximum has been applied to each female Luer cavity.

Leakage. The stopcock set shall be tested for leakage initially with the rubber injection site assembled and then with the rubber injection site removed. The end protectors of the set shall be removed and the rubber injection site shall be left intact.

Using test fittings in accordance with ISO Standard 594 each port shall be tested hydrostatically at a pressure of 45 psi for a period of 30 seconds.

The connection between the rotating spigot and the stopcock body shall be tested in the same manner except that a pressure of 20 psi shall be applied for a period of 30 seconds.

As an alternate, the testing may consist of an pneumatic circuit equivalent to the one above except that air shall be applied at a pressure of 10 psi. Any leakage shall be a test failure.

Leakage exceeding one clinging water drop shall be cause for failure. The rubber injection site shall then be removed and the ports tested in the same manner as above except that the female Luer cavity under the injection site shall be plugged.

Elevated Temperature. Samples of the packaged set shall be placed into an oven for seven days at 145° ± 2°F. The set shall then be observed for signs of deterioration.



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Security of assembly. The sample set shall be subjected to the security of assembly test after completion of the elevated temperature test.

The security of the stopcock injection site is to be tested for blow off. The stopcock shall be tested with an appropriate testing fluid applied internally to the stopcock. One of the female ports of the stopcock shall be tested by attaching the female port to a fitting in accordance with ISO Standard 594. The tubing port and the opposite female port are to be plugged. A continuous increasing fluid pressure is applied until the injection site is blown off or leaks fluid. The injection site must hold a minimum pressure of 100 psi before blow off.

The security of the tubing connections shall be tested. The connection being tested is to be held and a continuous increasing pull force applied to the tubing until the tubing connection separates. A minimum force of 10 lbs. is required before separation.

The security between the stopcock body and rotating spigot shall be tested. The stopcock body is to be held and a continuous increasing pull force applied to the rotating spigot until the rotating spigot separates from the body. A minimum force of 5 lbs. is required before separation.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.



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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

NOTE 1: The following National Stock Numbers are covered by this document:

| NSN | Item Identification | STYLE |
|---|---|---|
| 6515-00-864-8864 | STOPCOCK, Intravenous Therapy, Disposable, 50s. | 1 |
| 6515-01-307-3946 | STOPCOCK, Intravenous Therapy, 3-Way, with 33 Inch Extension Tube, Plastic, Disposable, 50s. | 2 |

NOTE 2: Copies of ISO Standards 594/1 and 594-2 may be obtained through the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.



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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-5398



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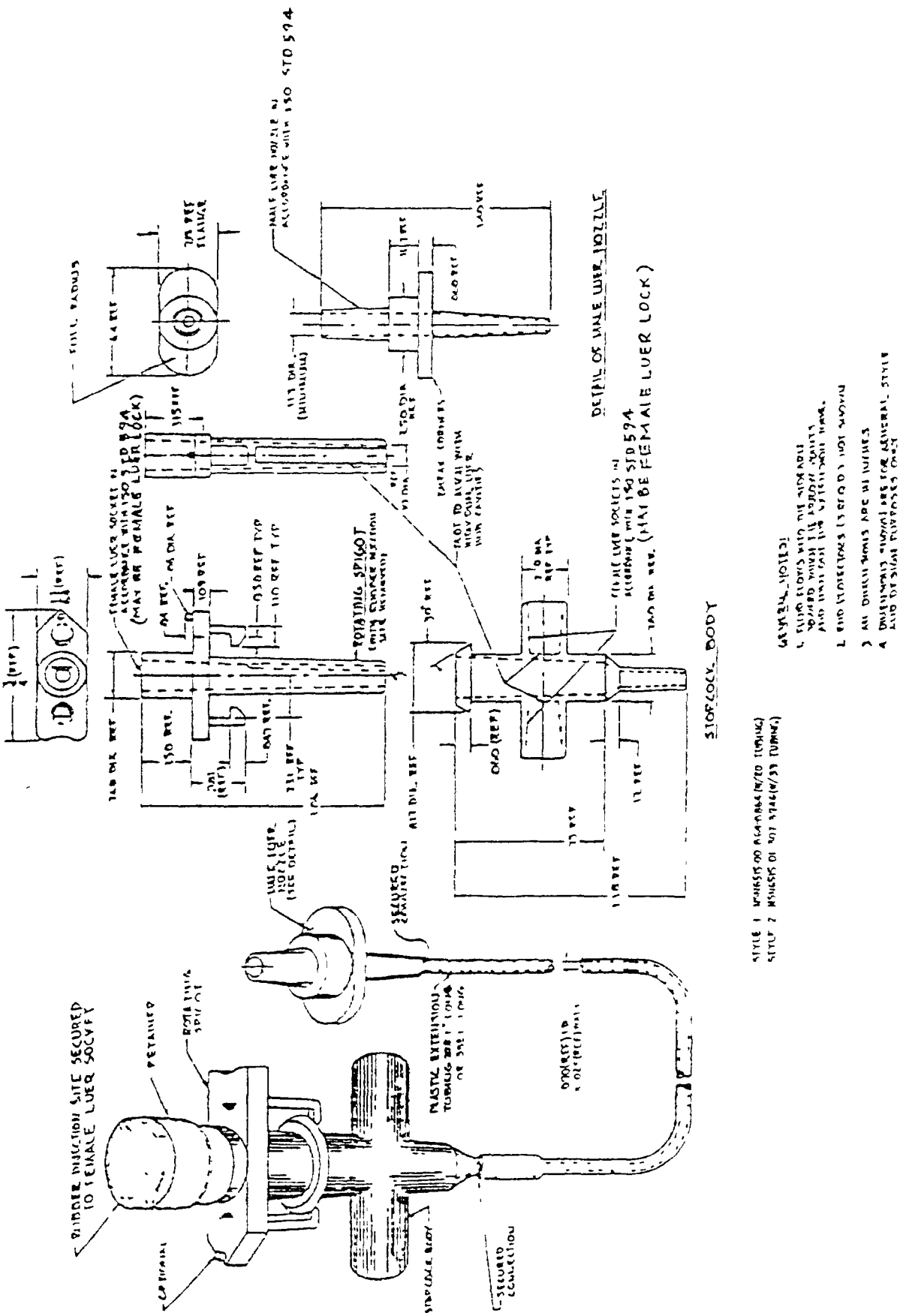
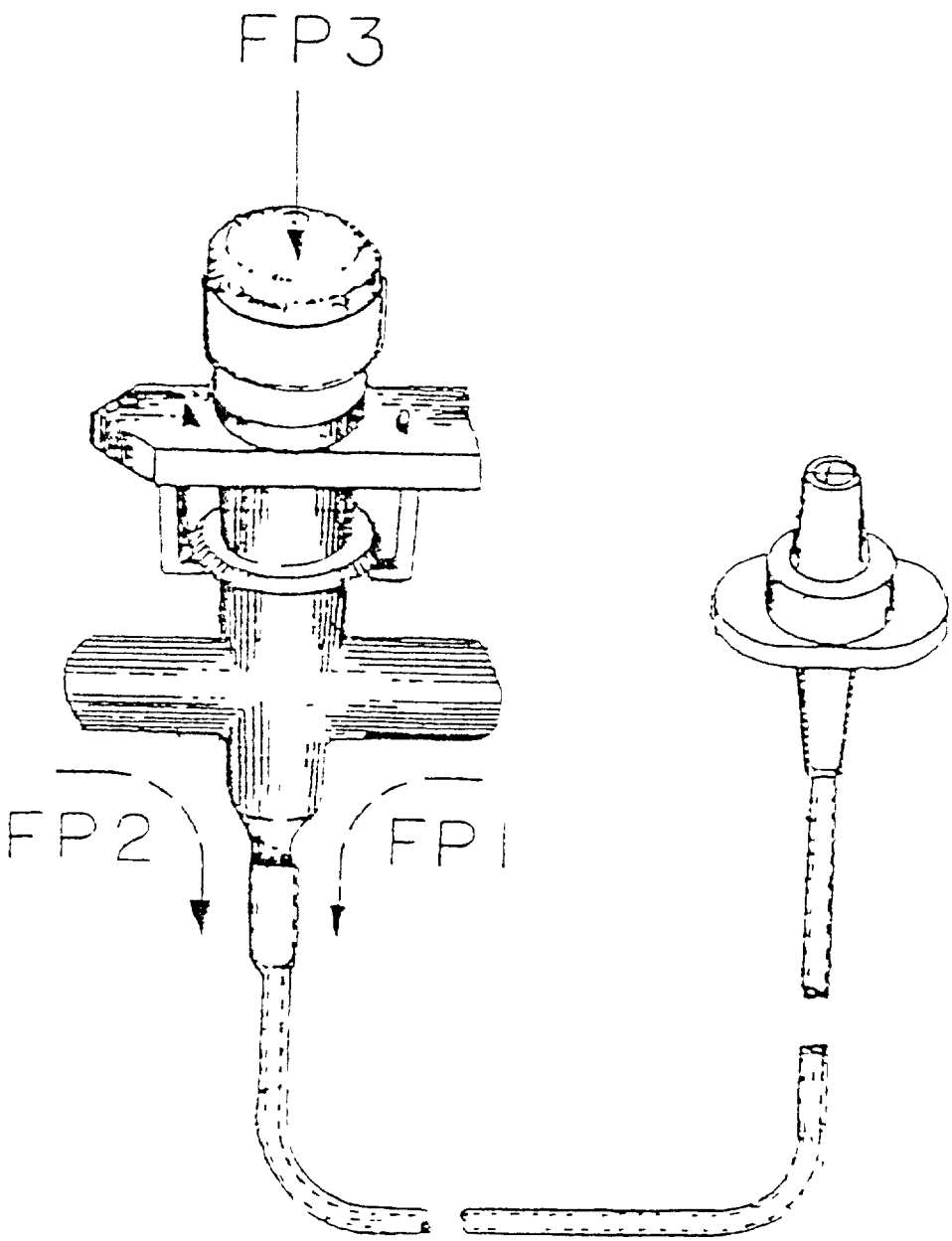



FIGURE 1 STOPCOCK, INTRAVENOUS THERAPY

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NSN 6515-00-864-8864

FIGURE 2 FLUID PATHS OF STOPCOCK



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

| I RECOMMEND A CHANGE: | 1 DOCUMENT NUMBER A-A-54746 | 2 DOCUMENT DATE (YYMMDD) 25 September 1992 |
|---|---|---|

3 DOCUMENT TITLE
STOPCOCK, INTRAVENOUS THERAPY, 3-WAY, PLASTIC

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

| a NAME (Last, First, Middle Initial) | b ORGANIZATION |
|---|---|
| c ADDRESS (Include Zip Code) | d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) / 7 DATE SUBMITTED (YYMMDD) |

8 PREPARING ACTIVITY

| a NAME | b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON |
|---|---|
| c ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340 |



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